

Owen Williamson

(919) 200-5910 • owenwilliamson997@gmail.com • www.linkedin.com/in/owen-k-williamson • https://github.com/owilli38

Professional Summary: M.S. in Data Science & Business Analytics (GPA: 3.90) with experience building machine learning models, NLP pipelines, and analytics dashboards. Skilled in Python, R, SQL, and Power BI with experience applying statistical and ML methods to healthcare and business problems.

TECHNICAL SKILLS

- **Languages:** R, Python, SQL
- **Tools:** Power BI, Excel, Supabase, Alteryx, Microsoft Suite, Google Workspace, Github
- **AI & Statistics:** RAG Pipelines, Prompt Engineering, Double Machine Learning, Causal Inference
- **Machine Learning:** Regression, Random Forest, XGBoost, HDBSCAN, K-Means, Time Series, NLP, Feature Engineering

WORK EXPERIENCE

Atrium Health/Wake Forest University

Winston-Salem, NC

Biostatistics Intern

June 2026 - Present

- Automated matching pipeline in R using the rentrez package to programmatically query PubMed for downstream publications for 1,650 institutional service forms.
- Extracted high-value keywords from unstructured text fields using TF-IDF text mining to identify related publications.
- Enhanced predictions of publication status using logistic regression with cross-validation to optimize tuning parameters.
- Analyzed the causal effect of institutional support on bibliometric measures using mixed-effect and zero-inflated linear models with fixed and random effect variables.
- Conducted systematic clinical chart reviews of Electronic Health Record (EHR) data to obtain high-fidelity ground truth variables and identify data quality gaps within structured clinical fields.

Fleet Feet

Huntersville, NC

Marketing Coordinator

March 2024 - May 2026

- Segmented a CRM database of 30,000+ contacts for targeted campaigns, increasing participation by 50%+.
- Streamlined operations by building an automated Excel KPI dashboard, reducing manual reporting constraints by 2 hours/week.
- Conducted SEO analysis to optimize web presence, successfully driving measurable increases in digital and foot traffic.
- Monitored social media engagement metrics through iterative testing, refining content strategy to boost engagement by 75%

Outfitter

August 2023 - May 2026

Center for Civic Engagement

Davidson, NC

Community Research Fellow

May 2023 - July 2023

- Automated ingestion and cleaning of multi-source education, census, geographic, and voter registration datasets using APIs.
- Built interactive dashboard visualizing disparities between school demographics and board representation across North Carolina.

PROJECTS

Pickaxe Analytics: State of Fraud Dashboard

January 2026 - May 2026

- Led analytical development of an end-to-end fraud intelligence pipeline within a five-person team.
- Built an automated scraping and NLP pipeline using embeddings and HDBSCAN to identify emerging fraud trends.
- Developed time-based metrics and a RAG pipeline to deliver evidence-based insights through an interactive dashboard.

Predicting Measles Outbreak Risk Across U.S. Counties

March 2026 - April 2026

- Built Logistic Regression, Random Forest, and XGBoost models to predict county-level measles outbreak risk.
- Engineered features from demographic and vaccination datasets.
- Achieved ROC-AUC of ~0.75 while optimizing recall for rare outbreak detection.

Unlock Hidden Revenue: Aligning Inventory to Optimize Sales

September 2025 - December 2025

- Applied time series forecasting to retail data to identify declining sales trends and uncover a \$25K+ annual revenue loss
- Designed an interactive Power BI dashboard to visualize supply-demand mismatches, enabling proactive inventory reallocation

EDUCATION

University of North Carolina, Charlotte

Charlotte, NC

Master of Science, Data Science and Business Analytics

August 2024 - May 2026

- GPA: 3.90
- Coursework: Statistical Learning; Applied Machine Learning; Econometrics; Database Systems

Davidson College

Davidson, NC

Bachelor of Arts, Political Science

August 2020 - January 2024

- GPA: 3.84, Magna Cum Laude
- Coursework: Multivariable Calculus & Modeling; Intro to Public Health; Intro to Research Methods in R